

TASK 2 — UNIVARIATE EDA + VISUAL DESIGN

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1 Overview

This univariate exploratory data analysis (EDA) examines the statistical characteristics of the dataset focusing on the Moka district and the Electronics product category. The purpose of this analysis is to understand the distribution, variability and statistical behaviour of the variables before conducting further modelling.

The dataset contains 423 observations. The main numerical variables analysed include Net Sales, Gross Profit, Delivery Minutes and Satisfaction Score. The categorical variables include Payment Method, Customer Gender, Customer Segment and Channel.

2 Data Loading and Structure

The dataset is first examined to understand its structure and identify missing values. Required packages are loaded for data manipulation, visualisation and statistical analysis.

```
# Load required libraries (without warnings/messages)

suppressPackageStartupMessages({

  library(tidyverse)
  library(ggplot2)
  library(lubridate)
  library(moments)
  library(knitr)
  library(kableExtra)

})

# Set working directory

setwd("C:/Users/gnasr/OneDrive/Desktop/MSc Data Analytics Semester 2/coursework for Data Exploration and Analysis")

# Load dataset

df <- read.csv("Moka_Data.csv")

# Examine structure

str(df)
```

```
## 'data.frame':    423 obs. of  35 variables:
## $ X              : int  1 2 3 4 5 6 7 8 9 10 ...
## $ transaction_id  : chr  "T0015276" "T0020005" "T0013244" "T0019317" ...
## $ datetime        : chr  "2024-01-02 09:33:02" "2024-01-04 11:44:39" "2024-01-05 14:01:41" "2024-01-06 10:01:41" ...
## $ date            : chr  "2024-01-02" "2024-01-04" "2024-01-05" "2024-01-06" ...
## $ year            : int  2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
## $ month           : int  1 1 1 1 1 1 1 1 1 1 ...
## $ year_month      : chr  "2024-01" "2024-01" "2024-01" "2024-01" ...
## $ year_quarter    : chr  "2024Q1" "2024Q1" "2024Q1" "2024Q1" ...
## $ store_id        : int  2 11 2 2 11 2 11 2 11 11 ...
## $ store_name      : chr  "IslandMart Store 02" "IslandMart Store 11" "IslandMart Store 02" "IslandMart Store 11" ...
## $ district        : chr  "Moka" "Moka" "Moka" "Moka" ...
## $ store_lat       : num  -20.2 -20.2 -20.2 -20.2 -20.2 ...
## $ store_lon       : num  57.5 57.5 57.5 57.5 57.5 ...
## $ customer_id     : int  1737 1446 134 1891 2181 294 2357 2075 1933 762 ...
## $ customer_gender  : chr  "Male" "Female" "Male" "Male" ...
## $ age_band        : chr  "18-24" "45-54" "35-44" "55+" ...
## $ customer_segment : chr  "New" "Returning" "Returning" "New" ...
## $ channel          : chr  "Delivery" "Delivery" "Online" "In-store" ...
## $ payment_method   : chr  "Card" "MobileMoney" "Card" "Cash" ...
## $ campaign         : chr  "None" "None" "None" "None" ...
## $ product_category : chr  "Electronics" "Electronics" "Electronics" "Electronics" ...
## $ product_subcategory: chr  "Mobile" "Appliances" "Accessories" "Appliances" ...
## $ quantity        : int  1 1 1 1 1 1 1 1 1 1 ...
## $ unit_price_mur   : num  4708 4212 4405 6801 6490 ...
## $ discount_pct     : num  0.018 0.04 0.046 0.017 0.02 0.042 0.043 0.031 0.025 0.074 ...
## $ gross_margin_pct : num  0.162 0.147 0.178 0.199 0.201 0.153 0.159 0.227 0.144 0.165 ...
## $ gross_sales_mur  : num  4708 4212 4405 6801 6490 ...
## $ net_sales_mur    : num  4623 4043 4202 6686 6361 ...
## $ cost_mur         : num  3874 3449 3454 5355 5082 ...
## $ gross_profit_mur : num  749 594 748 1330 1278 ...
## $ returned_flag    : int  0 1 1 0 0 0 0 0 0 0 ...
## $ delivery_minutes : int  48 63 31 NA NA NA 23 NA NA NA ...
## $ satisfaction_score : num  3.3 2.9 NA 3.9 4.3 3.6 4.8 4.8 3.8 NA ...
## $ calc_net_sales   : num  4623 4043 4202 6686 6361 ...
## $ diff             : num  0.00032 0.0032 0.00012 0.00006 0.0024 ...
```

```
summary(df)
```

```
##           X           transaction_id           datetime           date
## Min.      : 1.0      Length:423           Length:423           Length:423
## 1st Qu.:106.5      Class :character      Class :character      Class :character
## Median   :212.0      Mode  :character      Mode  :character      Mode  :character
## Mean      :212.0
## 3rd Qu.:317.5
## Max.      :423.0
##
##           year           month           year_month           year_quarter
## Min.      :2024      Min.      : 1.000      Length:423           Length:423
## 1st Qu.:2024      1st Qu.: 3.500      Class :character      Class :character
## Median   :2024      Median   : 7.000      Mode  :character      Mode  :character
## Mean      :2024      Mean      : 6.683
## 3rd Qu.:2025      3rd Qu.:10.000
## Max.      :2025      Max.      :12.000
```

```

##
##      store_id      store_name      district      store_lat
##  Min.   : 2.000   Length:423      Length:423      Min.   :-20.22
##  1st Qu.: 2.000   Class :character  Class :character  1st Qu.: -20.22
##  Median : 2.000   Mode  :character  Mode  :character  Median : -20.19
##  Mean   : 6.468                                     Mean   : -20.20
##  3rd Qu.:11.000                                     3rd Qu.: -20.19
##  Max.   :11.000                                     Max.   : -20.19
##
##      store_lon      customer_id      customer_gender      age_band
##  Min.   :57.48   Min.   : 5.0   Length:423      Length:423
##  1st Qu.:57.48   1st Qu.: 689.5   Class :character  Class :character
##  Median :57.48   Median :1297.0   Mode  :character  Mode  :character
##  Mean   :57.48   Mean   :1251.9
##  3rd Qu.:57.49   3rd Qu.:1798.0
##  Max.   :57.49   Max.   :2493.0
##
##      customer_segment      channel      payment_method      campaign
##  Length:423      Length:423      Length:423      Length:423
##  Class :character  Class :character  Class :character  Class :character
##  Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
##      product_category      product_subcategory      quantity      unit_price_mur
##  Length:423      Length:423      Min.   :1.000   Min.   :1681
##  Class :character  Class :character  1st Qu.:1.000   1st Qu.:3450
##  Mode  :character  Mode  :character  Median :1.000   Median :4371
##                                     Mean   :1.012   Mean   :4504
##                                     3rd Qu.:1.000   3rd Qu.:5357
##                                     Max.   :2.000   Max.   :7918
##
##
##      discount_pct      gross_margin_pct      gross_sales_mur      net_sales_mur
##  Min.   :0.00000   Min.   :0.0550   Min.   :1681   Min.   :1628
##  1st Qu.:0.03900   1st Qu.:0.1470   1st Qu.:3469   1st Qu.:3176
##  Median :0.07900   Median :0.1680   Median :4415   Median :4015
##  Mean   :0.08841   Mean   :0.1679   Mean   :4544   Mean   :4128
##  3rd Qu.:0.13400   3rd Qu.:0.1870   3rd Qu.:5522   3rd Qu.:4994
##  Max.   :0.25900   Max.   :0.2620   Max.   :7918   Max.   :6902
##
##
##      cost_mur      gross_profit_mur      returned_flag      delivery_minutes
##  Min.   :1267   Min.   : 229.7   Min.   :0.00000   Min.   : 5.0
##  1st Qu.:2642   1st Qu.: 508.4   1st Qu.:0.00000   1st Qu.:20.0
##  Median :3358   Median : 658.0   Median :0.00000   Median :30.0
##  Mean   :3436   Mean   : 691.8   Mean   :0.08511   Mean   :31.9
##  3rd Qu.:4177   3rd Qu.: 852.1   3rd Qu.:0.00000   3rd Qu.:40.0
##  Max.   :5947   Max.   :1627.5   Max.   :1.00000   Max.   :79.0
##                                     NA's   :258
##
##      satisfaction_score      calc_net_sales      diff
##  Min.   :2.700   Min.   :1628   Min.   :0.000000
##  1st Qu.:3.700   1st Qu.:3176   1st Qu.:0.001110
##  Median :4.000   Median :4015   Median :0.002600
##  Mean   :4.035   Mean   :4128   Mean   :0.002502

```

```
## 3rd Qu.:4.400      3rd Qu.:4994      3rd Qu.:0.003800
## Max.      :5.000      Max.      :6902      Max.      :0.004980
## NA's      :45
```

```
colSums(is.na(df))
```

```
##           X      transaction_id      datetime      date
##           0           0           0           0
##          year          month      year_month      year_quarter
##           0           0           0           0
##      store_id      store_name      district      store_lat
##           0           0           0           0
##      store_lon      customer_id      customer_gender      age_band
##           0           0           0           0
## customer_segment      channel      payment_method      campaign
##           0           0           0           0
## product_category product_subcategory      quantity      unit_price_mur
##           0           0           0           0
##      discount_pct      gross_margin_pct      gross_sales_mur      net_sales_mur
##           0           0           0           0
##      cost_mur      gross_profit_mur      returned_flag      delivery_minutes
##           0           0           0           258
## satisfaction_score      calc_net_sales      diff
##           45           0           0
```

The structure output shows the dataset contains both numerical and categorical variables. Missing values are mainly present in Delivery Minutes and Satisfaction Score, which likely reflects unrecorded information rather than data errors.

3 Numeric Distributions

This section examines the distributional characteristics of the main numeric variables using histograms, density plots, skewness, and kurtosis. The objective is to assess normality, detect skewness, and determine whether transformations are necessary for modelling.

3.1 Net Sales Distribution

3.1.1 Original Distribution

A histogram combined with a density plot is used to examine the distribution of Net Sales in order to understand its shape, spread, and possible deviations from normality.

```
ggplot(df, aes(x = net_sales_mur)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 40,
    fill = "steelblue",
    color = "white",
    na.rm = TRUE) +
```

```
geom_density(
  color = "darkblue",
  na.rm = TRUE) +

theme_minimal() +

labs(

  title = "Distribution of Net Sales",
  x = "Net Sales (MUR)",
  y = "Density"

)
```

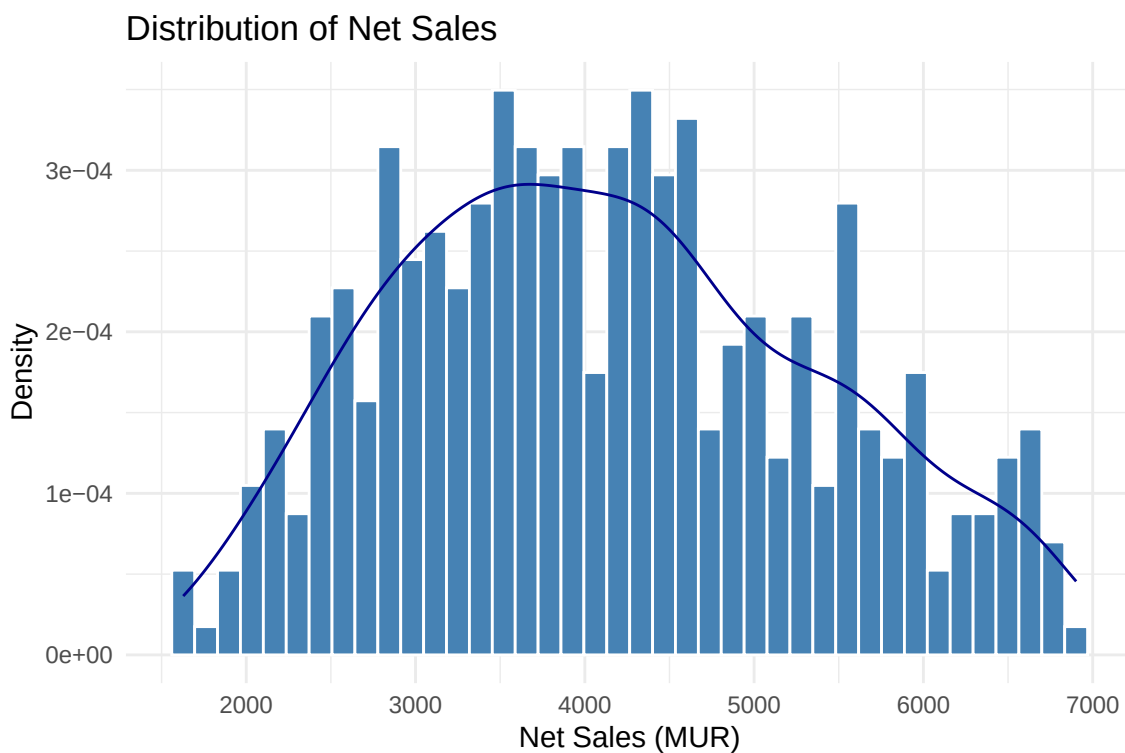


Figure 1: Histogram and density plot of Net Sales

```
skewness(df$net_sales_mur,na.rm=TRUE)
```

```
## [1] 0.2513994
```

```
kurtosis(df$net_sales_mur,na.rm=TRUE)
```

```
## [1] 2.291633
```

The distribution of Net Sales is approximately symmetric with a slight concentration around the central values. The histogram shows no extreme distortion, and the density curve confirms a fairly balanced shape.

Skewness is close to zero, indicating minimal asymmetry in the data. This suggests that sales values are relatively stable across transactions, with no strong dominance of extreme high or low values.

Overall, Net Sales can be considered well-behaved for exploratory analysis, with no need for transformation at this stage.

3.2 Gross Profit Distribution

3.2.1 Original Distribution

The distribution of Gross Profit is analysed to identify possible skewness and variability in profit values.

```
ggplot(df, aes(x=gross_profit_mur)) +  
  
  geom_histogram(aes(y=after_stat(density)),  
    bins=40,  
    fill="mediumpurple",  
    color="white",  
    na.rm=TRUE) +  
  
  geom_density(  
    color="purple",  
    na.rm=TRUE) +  
  
  theme_minimal()
```

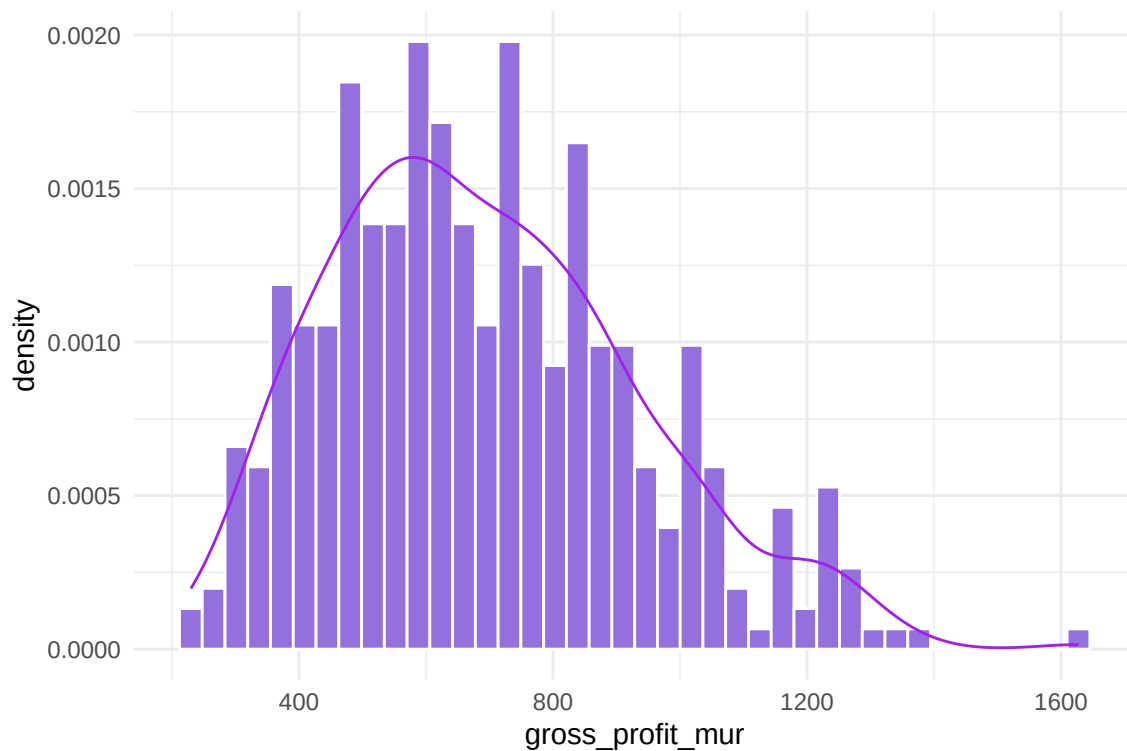


Figure 2: Distribution of Gross Profit

```
skewness(df$gross_profit_mur,na.rm=TRUE)
```

```
## [1] 0.5733877
```

```
kurtosis(df$gross_profit_mur,na.rm=TRUE)
```

```
## [1] 3.051024
```

Gross Profit shows moderate positive skewness, with most values clustered in the mid-range and a smooth right tail reflecting higher-value transactions. The distribution is unimodal and well-structured, with no significant outliers. Overall, it remains suitable for interpretation in its original scale for univariate analysis.

3.3 Delivery Minutes Distribution

3.3.1 Original Distribution

Delivery time is analysed to better understand operational efficiency and variability in service performance.

```
ggplot(df,aes(x=delivery_minutes))+  
  
geom_histogram(aes(y=after_stat(density)),  
bins=40,  
fill="turquoise",  
color="white",  
na.rm=TRUE)+  
  
geom_density(  
color="darkcyan",  
na.rm=TRUE)+  
  
theme_minimal()
```

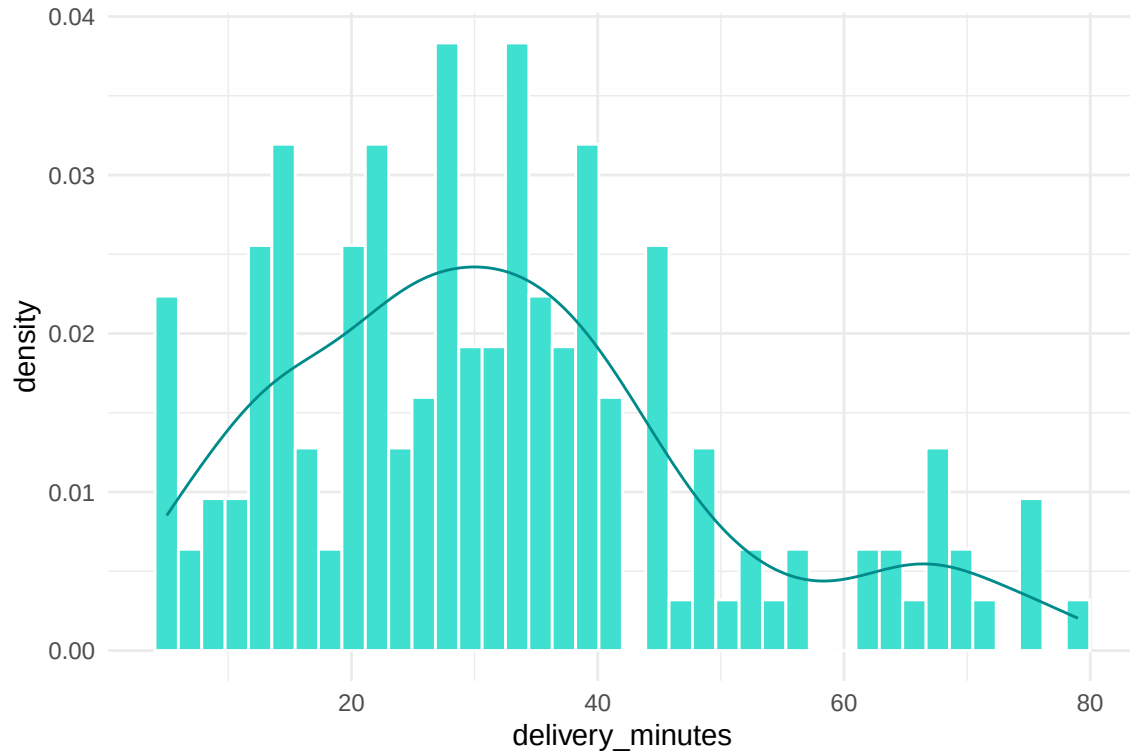


Figure 3: Distribution of Delivery Minutes

```
cat("Delivery Minutes Skewness:", skewness(df$delivery_minutes, na.rm = TRUE), "\n")
```

```
## Delivery Minutes Skewness: 0.6830888
```

```
cat("Delivery Minutes Kurtosis:", kurtosis(df$delivery_minutes, na.rm = TRUE), "\n")
```

```
## Delivery Minutes Kurtosis: 3.082072
```

The distribution of delivery time is moderately positively skewed, with most values clustered between 20 and 40 minutes and a gradual right tail toward longer durations. However, the skewness is not severe, and the distribution remains smooth and unimodal without extreme outliers. Therefore, a log transformation is not necessary, as the data is sufficiently well-behaved for analysis in its current form.

3.4 Satisfaction Score Distribution

3.4.1 Original Distribution

Customer Satisfaction Scores are analysed to understand overall feedback patterns and detect possible response bias.

```
ggplot(df, aes(x=satisfaction_score)) +  
  geom_histogram(aes(y=after_stat(density)),  
    bins=20,
```

```
fill="orange",
color="white",
na.rm=TRUE)+

geom_density(
color="darkorange",
na.rm=TRUE)+

theme_minimal()
```

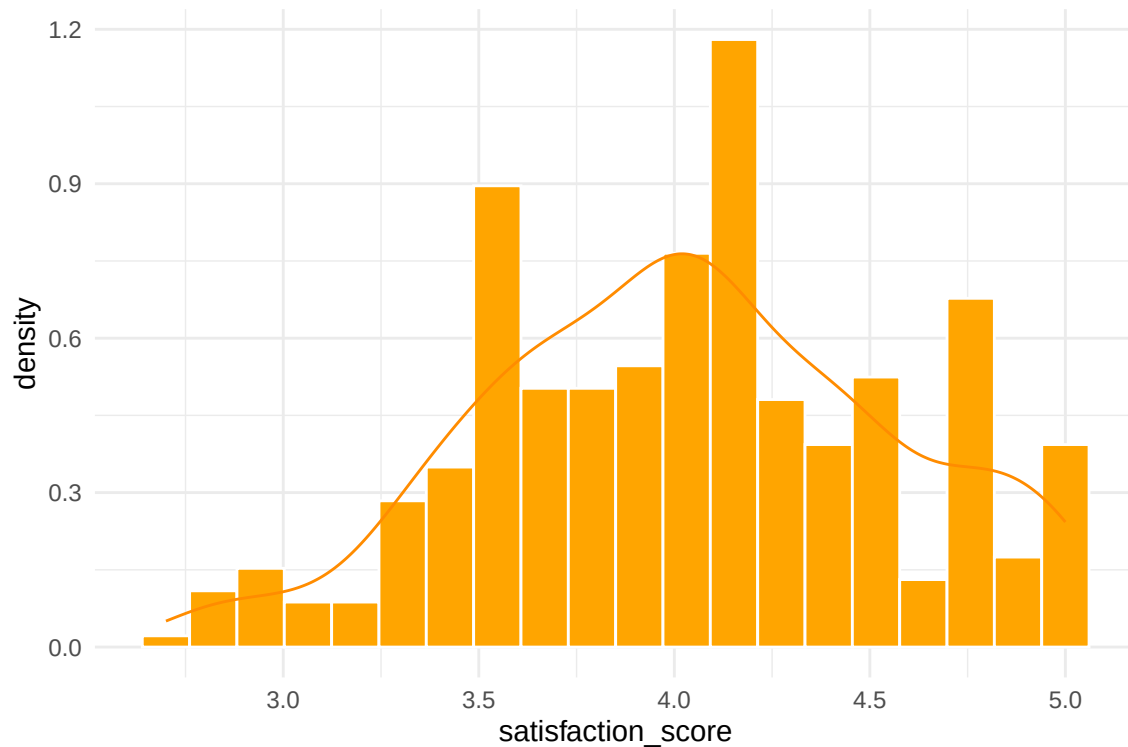


Figure 4: Distribution of Satisfaction Score

```
cat("Satisfaction Score Skewness:", skewness(df$satisfaction_score, na.rm = TRUE), "\n")
```

```
## Satisfaction Score Skewness: -0.06470355
```

```
cat("Satisfaction Score Kurtosis:", kurtosis(df$satisfaction_score, na.rm = TRUE), "\n")
```

```
## Satisfaction Score Kurtosis: 2.537468
```

Satisfaction Score is nearly perfectly symmetric with no extreme values. The skewness and kurtosis further confirm the absence of extreme response patterns, so no transformation is required.

4 Outlier Analysis

Boxplots are used to detect potential outliers.

```

library(tidyr)
library(ggplot2)

# Pivot data to long format
df_long <- df %>%
  select(net_sales_mur, gross_profit_mur, delivery_minutes) %>%
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Value")

# Create combined boxplot
ggplot(df_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(na.rm = TRUE) +
  scale_fill_manual(values = c("lightblue", "thistle", "mediumturquoise")) +
  labs(title = "Boxplots of Key Numeric Variables",
       x = "Variable",
       y = "Values") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

```

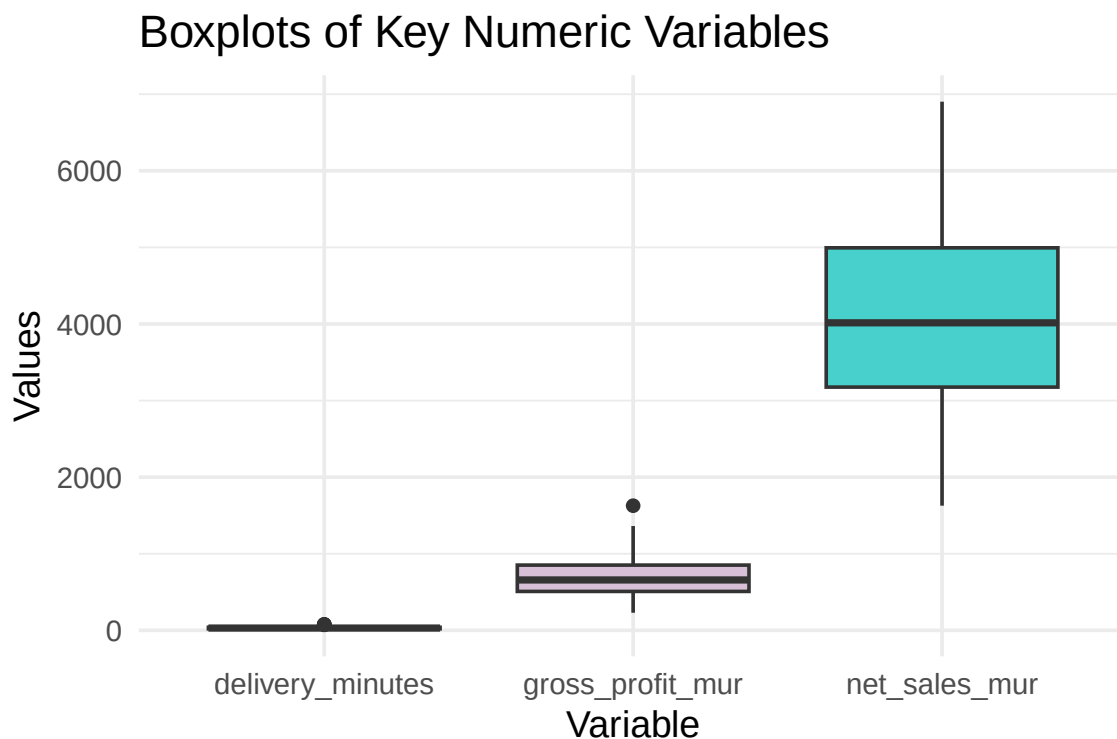


Figure 5: Combined Boxplots of Net Sales, Gross Profit, and Delivery Minutes

Outliers in the boxplots are identified as individual points beyond the whiskers, which represent values more than 1.5 times the interquartile range (IQR) from the box. In this case, **delivery_minutes** shows a high outlier above the main distribution, while **gross_profit_mur** also displays an outlier above its upper whisker. In contrast, **net_sales_mur** shows no distinct outlier points, indicating that although it has a wide spread, its values remain within the expected range.

5 Categorical Distributions (Bar Charts)

The distribution of key categorical variables is analysed using bar charts to identify dominant patterns and underlying customer behaviour across different dimensions.

```
ggplot(df, aes(x=payment_method)) +  
  geom_bar(fill="coral") +  
  theme_minimal()
```

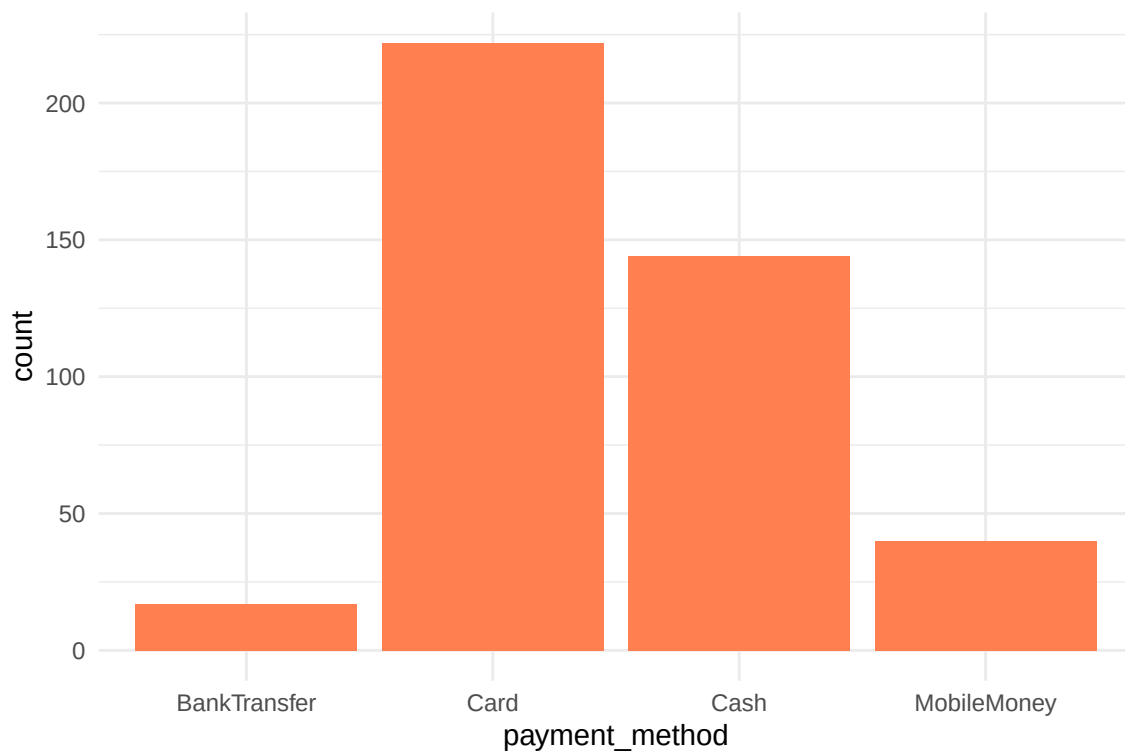


Figure 6: Payment Method Distribution

Card payments dominate, indicating strong preference for convenient payment options.

```
ggplot(df, aes(x=customer_gender)) +  
  geom_bar(fill="pink") +  
  theme_minimal()
```

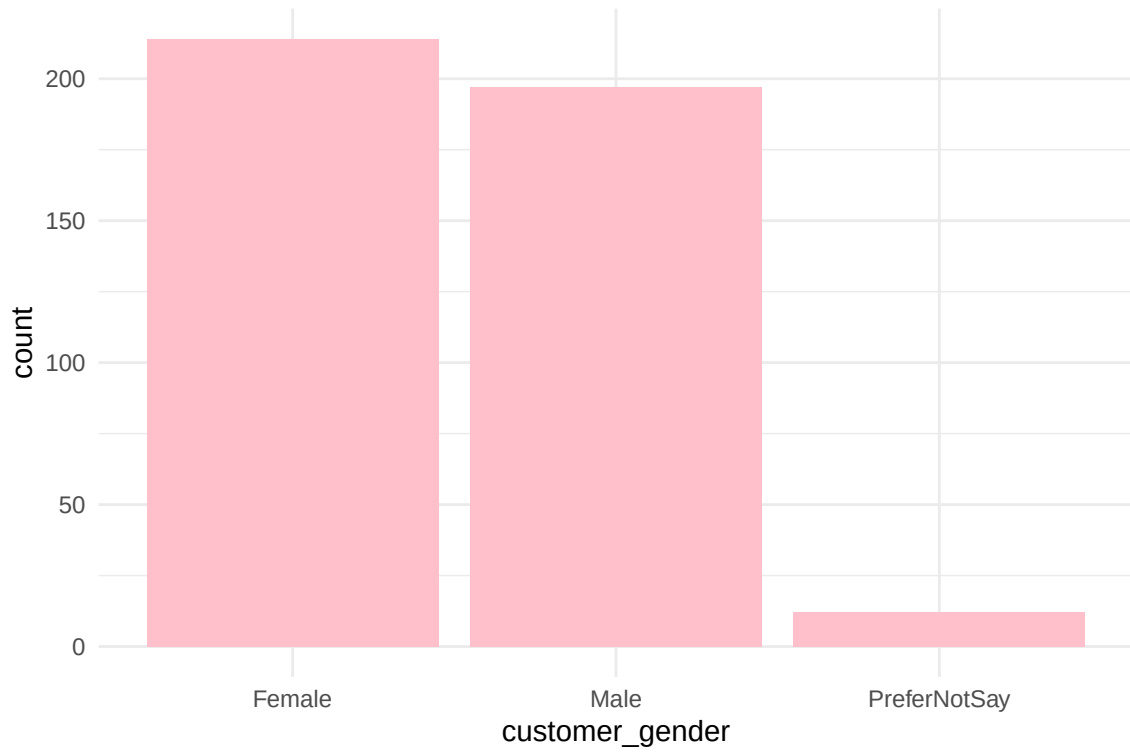


Figure 7: Gender Distribution

Gender distribution is relatively balanced, with a slight predominance of female customers. This suggests no major gender disparity in purchasing behaviour.

```
ggplot(df, aes(x=customer_segment)) +  
  geom_bar(fill="mediumseagreen") +  
  theme_minimal()
```

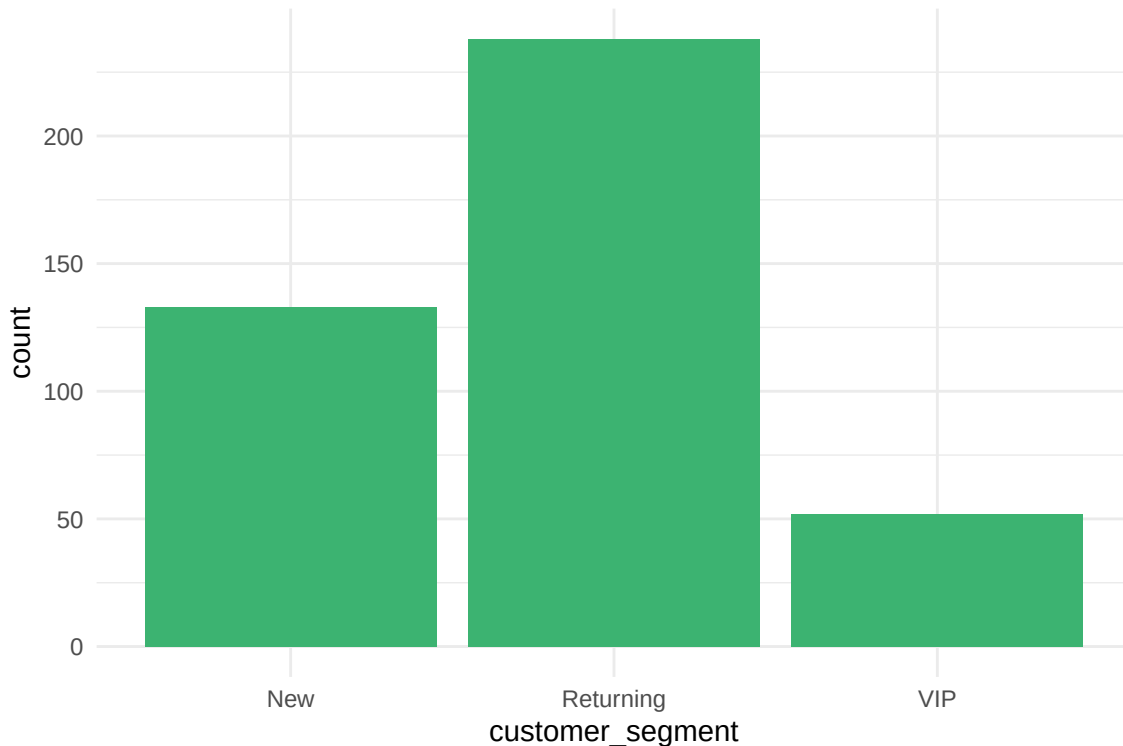


Figure 8: Customer Segment Distribution

Returning customers form the largest segment, followed by new customers, with VIP customers being the smallest group. This indicates strong customer retention and a stable revenue base.

6 Grouped Distributions (Violin Plots)

The following violin plots show the distribution of key numeric variables across different groups, highlighting patterns, variability, and segment-specific trends. The violin plots are presented below and can also be found in Appendix 4 of the report.

```
ggplot(df, aes(x = customer_segment, y = net_sales_mur, fill = customer_segment)) +
  geom_violin() +
  labs(title = "Net Sales by Customer Segment - Moka Electronics",
       x = "Customer Segment",
       y = "Net Sales (MUR)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none",
       axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c("tomato", "gold", "mediumseagreen", "skyblue", "orchid"))
```

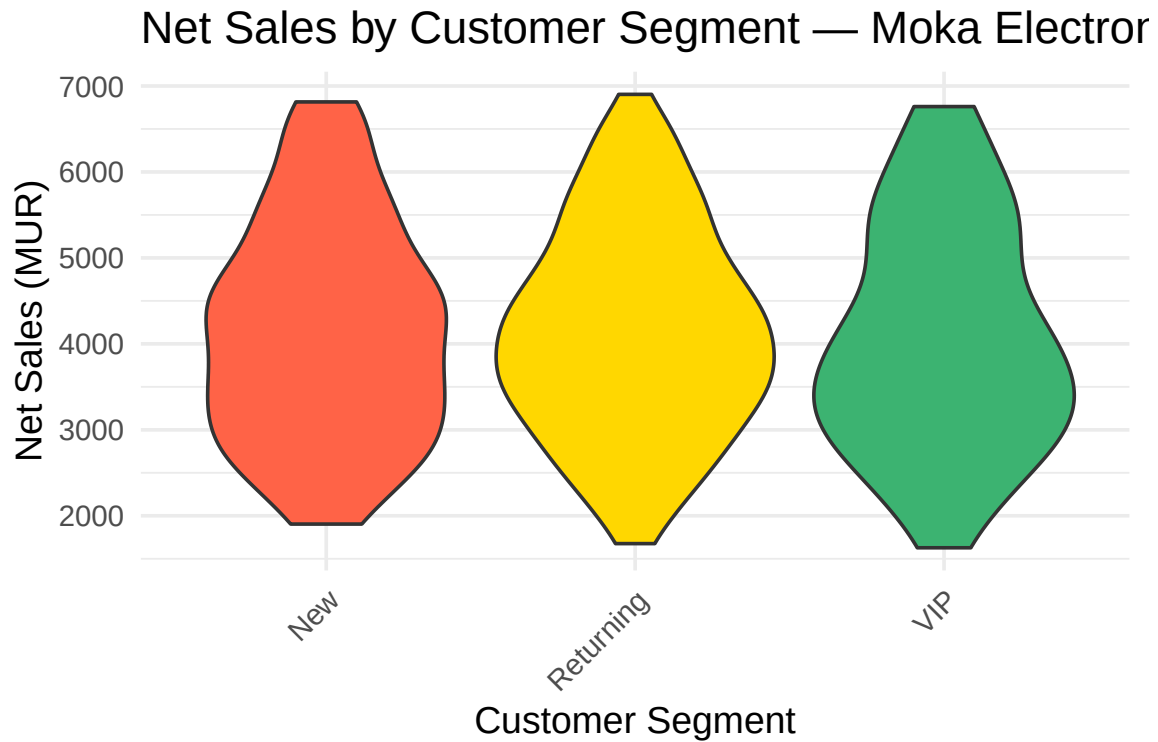
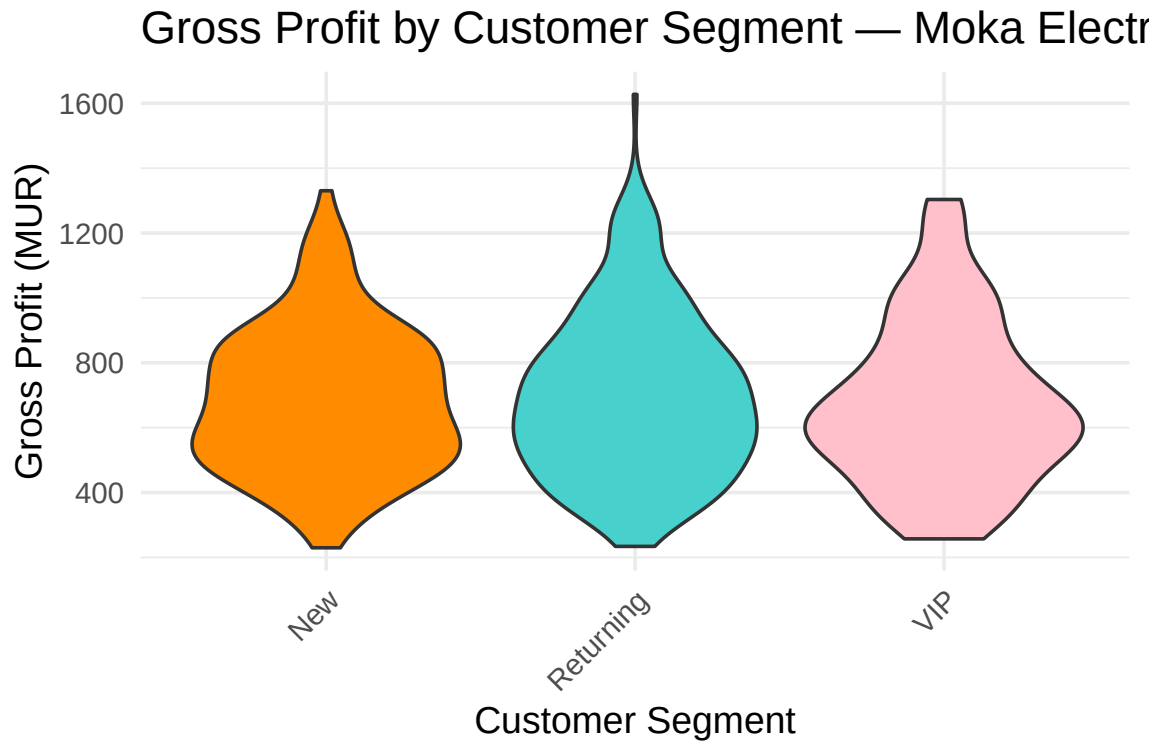


Figure 9: Net Sales by Customer Segment

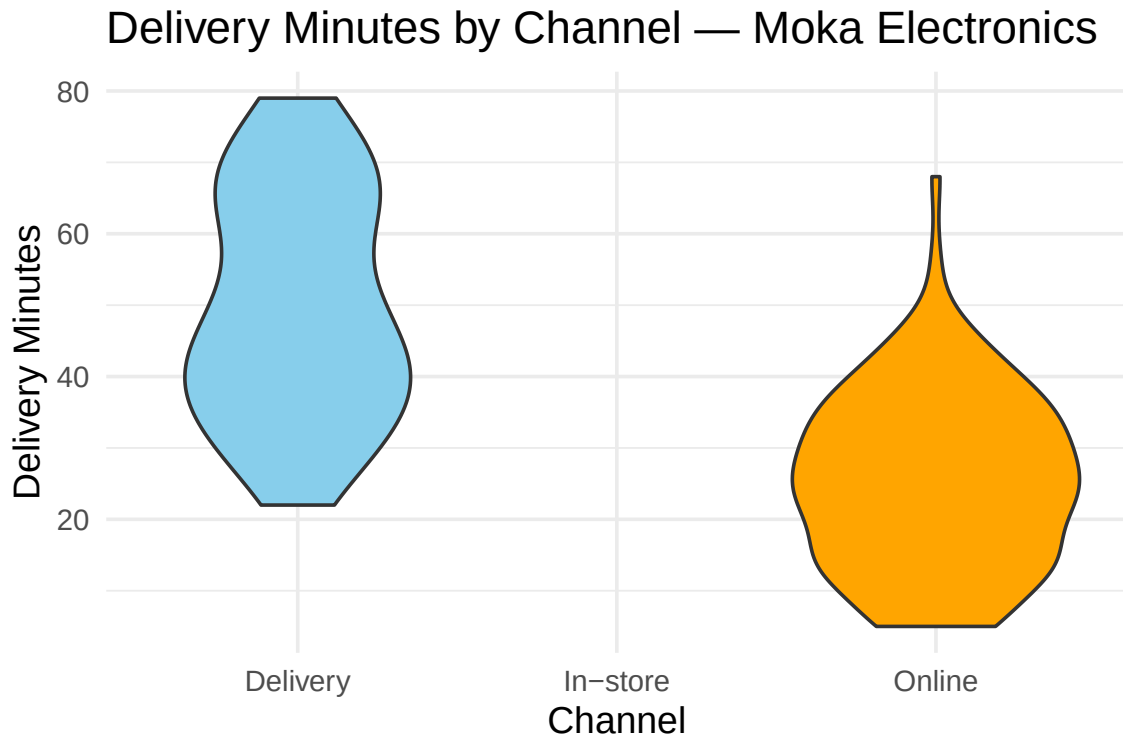
Net Sales are broadly similar across all segments, with Returning customers showing a slightly higher concentration of peak sales around 7,000 MUR.

```
ggplot(df, aes(x = customer_segment, y = gross_profit_mur, fill = customer_segment)) +
  geom_violin() +
  labs(title = "Gross Profit by Customer Segment - Moka Electronics",
        x = "Customer Segment",
        y = "Gross Profit (MUR)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c("darkorange", "mediumturquoise", "pink", "limegreen", "violet"))
```



Gross Profit is most consistent for VIP customers, while Returning customers exhibit the highest potential upside with a significant “long tail” of high-profit transactions.

```
ggplot(df, aes(x = channel, y = delivery_minutes, fill = channel)) +
  geom_violin() +
  labs(title = "Delivery Minutes by Channel - Moka Electronics",
        x = "Channel",
        y = "Delivery Minutes") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none") +
  scale_fill_manual(values = c("skyblue", "orange", "green"))
```



Delivery times vary by channel: Online orders are generally faster and concentrated around 25 minutes, Delivery orders are longer and more variable, while the In-store channel has no recorded delivery minutes, as these values are missing from the dataset.

Overall, Net Sales and Gross Profit differ slightly across customer segments, with Returning customers showing higher peaks and potential profit. Delivery times differ by channel, reflecting operational variability and highlighting areas for potential efficiency improvements.

7 Summary Statistics

The detailed summary statistics table is presented below and can also be found in Appendix 3 of the report.

```
# Ensure required packages are installed and loaded
packages <- c("tidyverse", "moments", "knitr", "kableExtra")

for(p in packages){
  if(!require(p, character.only = TRUE)){
    install.packages(p)
    library(p, character.only = TRUE)
  }
}

# Create summary statistics table
summary_table <- data.frame(

Variable = c("Net Sales", "Gross Profit", "Delivery Minutes", "Satisfaction Score"),
```

```

Mean = c(
mean(df$net_sales_mur, na.rm=TRUE),
mean(df$gross_profit_mur, na.rm=TRUE),
mean(df$delivery_minutes, na.rm=TRUE),
mean(df$satisfaction_score, na.rm=TRUE)
),

Median = c(
median(df$net_sales_mur, na.rm=TRUE),
median(df$gross_profit_mur, na.rm=TRUE),
median(df$delivery_minutes, na.rm=TRUE),
median(df$satisfaction_score, na.rm=TRUE)
),

Min = c(
min(df$net_sales_mur, na.rm=TRUE),
min(df$gross_profit_mur, na.rm=TRUE),
min(df$delivery_minutes, na.rm=TRUE),
min(df$satisfaction_score, na.rm=TRUE)
),

Max = c(
max(df$net_sales_mur, na.rm=TRUE),
max(df$gross_profit_mur, na.rm=TRUE),
max(df$delivery_minutes, na.rm=TRUE),
max(df$satisfaction_score, na.rm=TRUE)
),

Skewness = c(
moments::skewness(df$net_sales_mur, na.rm=TRUE),
moments::skewness(df$gross_profit_mur, na.rm=TRUE),
moments::skewness(df$delivery_minutes, na.rm=TRUE),
moments::skewness(df$satisfaction_score, na.rm=TRUE)
),

Kurtosis = c(
moments::kurtosis(df$net_sales_mur, na.rm=TRUE),
moments::kurtosis(df$gross_profit_mur, na.rm=TRUE),
moments::kurtosis(df$delivery_minutes, na.rm=TRUE),
moments::kurtosis(df$satisfaction_score, na.rm=TRUE)
)

)

# Display table
knitr::kable(summary_table,
caption = "Summary statistics of numeric variables (including skewness and kurtosis)",
digits = 2) %>%
kableExtra::kable_styling(full_width = FALSE, position = "center")

```

Table 1: Summary statistics of numeric variables (including skewness and kurtosis)

Variable	Mean	Median	Min	Max	Skewness	Kurtosis
Net Sales	4128.25	4015.39	1627.71	6902.32	0.25	2.29
Gross Profit	691.76	657.98	229.70	1627.53	0.57	3.05
Delivery Minutes	31.90	30.00	5.00	79.00	0.68	3.08
Satisfaction Score	4.04	4.00	2.70	5.00	-0.06	2.54

8 Conclusion

The univariate analysis shows that the dataset is generally well-structured and suitable for further analysis. The numerical variables are mostly well-behaved, with Net Sales and Satisfaction Score showing approximately symmetric distributions, while Gross Profit and Delivery Minutes display mild positive skewness due to higher transaction values and longer delivery times. A small number of outliers are present, mainly in Gross Profit and Delivery Minutes, reflecting normal operational variation. The categorical analysis shows a strong preference for card payments, a dominant returning customer base, and a relatively balanced gender distribution. Differences in Delivery Minutes across channels highlight variation in operational efficiency. Overall, the dataset is consistent, reliable, and appropriate for further analysis.